

Conformal Geodesic Cardiovascular Coils for Ricci-State Exploration

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Abstract

Cardiovascular MRI coil design has historically optimized signal-to-noise ratio (SNR) and coverage in Euclidean approximations of anatomy. This work introduces a conformal-geometric design framework in which receive elements are distributed along geodesics of a patient-specific cardiac manifold, while field uniformity and encoding sensitivity are regulated by Ricci-state curvature constraints. We define a finite computational pipeline that maps segmented cardiovascular anatomy to a conformal metric, derives geodesic coil trajectories, and enforces curvature-aware sensitivity regularization. In simulation, the proposed conformal geodesic architecture improves regional encoding efficiency in left ventricular and aortic root territories while reducing sensitivity drop-off at high-curvature interfaces. We further provide a translational engineering roadmap for manufacturable flexible arrays and discuss clinical implications for flow imaging, myocardial tissue characterization, and intervention guidance.

1. Introduction

Cardiovascular imaging presents a geometric paradox: the most diagnostically relevant structures are often those with high curvature, anisotropic motion, and rapidly changing local topology under cardiac and respiratory cycles. Standard coil geometries are typically designed on simplified surfaces and then adapted to patients. This sequence can produce sensitivity heterogeneity in regions where morphology is most informative.

Conformal geodesic coils invert this process. We first estimate a patient-aligned manifold and then place conductive paths and element centers along geodesics induced by a curvature-aware metric. Ricci-state exploration is used to quantify and control local distortion pressure in the manifold-to-coil mapping, improving robustness of sensitivity profiles in curved domains.

Contributions

1. A conformal metric formulation for cardiovascular coil layout optimization.

2. A Ricci-state energy term that controls local field distortion in high-curvature regions.
3. A finite discrete algorithm for geodesic coil path synthesis under manufacturability constraints.
4. A translational implementation strategy for flexible cardiovascular array fabrication.

2. Geometric framework

2.1 Cardiovascular manifold and conformal metric

Let $\mathcal{M}$ denote a triangulated cardiovascular surface (pericardial envelope plus vessel segments) derived from segmentation. A conformal metric is defined as:

$$[Equation: g_{ij}(x) = e^{2\phi(x)} \bar{g}_{ij}(x) \dots]$$

where $\bar{g}_{ij}$ is the baseline metric on $\mathcal{M}$ and $\phi(x)$ is a scalar conformal potential learned from anatomical priorities (e.g., myocardium, valves, aortic arch).

Conformal scaling preserves local angles while reweighting distance, allowing us to prioritize geodesic density in diagnostically critical regions without introducing discontinuous topological artifacts.

2.2 Geodesic element trajectories

A coil trace $\gamma(s)$ is computed by minimizing metric length:

$$[Equation: L[\gamma] = \int_0^1 \sqrt{g_{ij}(\gamma(s)) \dot{\gamma}^i(s) \dot{\gamma}^j(s)} ds \dots]$$

subject to endpoint and fabrication constraints (minimum bend radius, route separation, connector accessibility).

The resulting Euler-Lagrange equations define geodesic paths in the conformal manifold and are solved numerically by discrete geodesic shooting with projected constraints.

2.3 Ricci-state representation

We define Ricci-state scalar field $\rho(x)$ from discrete Ricci curvature:

[Equation: $\rho(x)=R(x)-\langle R \rangle_{\Omega}$...]

where $R(x)$ is local scalar curvature and $\langle R \rangle_{\Omega}$ is domain average over target region Ω .

High absolute $\rho(x)$ indicates local geometric stress likely to induce sensitivity distortion in naive layouts. We therefore include Ricci-state regularization in optimization.

3. Coil optimization objective

We optimize element centers, conductor paths, and tuning distribution by minimizing:

[Equation: $J = \alpha J_{\text{SNR}} + \beta J_{\text{hom}} + \gamma J_{\text{dec}} + \delta J_{\text{Ricci}} + \eta J_{\text{fab}}$...]

with terms:

- J_{SNR} : inverse regional SNR utility in target cardiovascular territories.
- J_{hom} : B1- field nonuniformity penalty.
- J_{dec} : inter-element coupling and g-factor proxy penalty.
- $J_{\text{Ricci}} = \int_{\Omega} |\nabla S(x)|^2 (1 + |\rho(x)|) dx$: curvature-weighted sensitivity roughness.
- J_{fab} : manufacturability penalties (bend radius, overlap, trace spacing, connector reachability).

The Ricci-weighted term increases smoothness pressure where curvature stress is high, preventing severe local sensitivity collapse.

4. Finite discrete implementation

4.1 Discretization

On mesh nodes v_k , the discrete energy is:

[Equation: $J_{\text{Ricci}}^{\text{disc}} = \sum_{(i,j) \in E} w_{ij} (S_i - S_j)^2 \left(1 + \frac{|\rho_i| + |\rho_j|}{2}\right)$...]

where E is edge set and w_{ij} are cotangent-style weights.

4.2 Iterative solver

Algorithmic cycle:

1. Segment anatomy and construct manifold mesh.
2. Estimate conformal potential ϕ from clinical priority map.
3. Initialize element anchors on low-strain geodesic seeds.
4. Solve constrained geodesic trajectories for conductor traces.
5. Update sensitivity model and coupling matrix.
6. Evaluate objective and apply projected quasi-Newton step.
7. Stop at tolerance or max iteration.

4.3 Computational complexity

With N_v mesh vertices, N_e elements, and K optimization iterations:

[Equation: $\mathcal{O}(K(N_v \log N_v + N_e^3)) \dots$]

where $N_v \log N_v$ stems from repeated geodesic solves and N_e^3 from coupling-aware linear algebra in tuning updates.

5. Cardiovascular simulation protocol

5.1 Anatomical targets

We evaluate three representative domains:

- Left ventricle and interventricular septum
- Aortic root and proximal ascending aorta

- Pulmonary outflow tract neighborhood

5.2 Baselines

1. Conventional cylindrical array projection
2. Uniformly sampled body-surface geodesic array without Ricci weighting
3. Proposed conformal geodesic + Ricci-state optimization

5.3 Metrics

- Regional SNR (median, 10th percentile)
- B1- coefficient of variation in target ROI
- Coupling proxy and conditioning score
- Distortion-sensitive sensitivity gradient index

6. Results (technical summary)

The conformal geodesic Ricci-state design demonstrates:

- Higher low-percentile SNR in high-curvature anatomical sectors.
- Improved sensitivity continuity near the aortic root transition.
- Reduced coupling penalties after curvature-aware spacing adjustments.
- Better robustness under moderate anatomical deformation perturbations.

The largest gains occur where classical Euclidean flattening induces coordinate distortion, supporting the geometric premise of the method.

7. Translational engineering considerations

7.1 Fabrication pathway

A practical route uses flexible multilayer substrates with segmented conductor islands linked by compliant interconnects. This supports geodesic trace fidelity while preserving patient comfort and connector reliability.

7.2 Tuning and safety

Ricci-aware element layouts require integrated co-optimization of:

- preamplifier decoupling networks,
- distributed capacitive tuning,
- SAR-constrained drive and receive profiles.

Curvature-prioritized zones should include conservative thermal guard bands and redundant monitoring points.

7.3 Clinical workflow

A deployable workflow can precompute patient-class templates (small, medium, large thoracic phenotypes) and refine only anchor positions online, reducing planning overhead.

8. Discussion

Conformal geodesic coil design reframes cardiovascular array engineering as a problem in differential geometry under practical device constraints. Ricci-state exploration offers a compact quantitative descriptor for where geometric stress is likely to degrade sensitivity smoothness and reconstruction conditioning.

This approach is not limited to cardiovascular systems; it can be extended to neurovascular and abdominal manifolds where anisotropy and curvature are similarly dominant. The key translational challenge is balancing geometric optimality against assembly complexity and regulatory validation burden.

9. Limitations and future work

1. Current simulations use static snapshots and do not fully resolve cycle-by-cycle moving-manifold effects.
2. Electromagnetic full-wave coupling is approximated in the optimization loop for computational tractability.
3. Clinical outcome linkage (diagnostic confidence, intervention planning accuracy) remains to be quantified prospectively.

Future work should integrate motion-aware Ricci flow updates, patient-specific uncertainty quantification, and closed-loop tuning control during acquisition.

10. Conclusion

We present a technical framework for cardiovascular coil design in which conformal geodesics define physically realizable array topology and Ricci-state exploration regularizes field behavior in curvature-critical regions. The resulting architecture improves encoding stability where conventional coil layouts are most vulnerable. These findings support conformal differential geometry as a practical next step in high-performance cardiovascular MRI hardware design.

Methods Appendix: Ricci-state constrained geodesic update

At iteration t , with element state vector θ_t :

$$\theta_{t+1} = \Pi_{\mathcal{C}}(\theta_t - \mu_t H_t^{-1} \nabla J(\theta_t)) \dots$$

where $\Pi_{\mathcal{C}}$ projects onto fabrication-feasible set $\mathcal{C}$ and H_t is a quasi-Newton approximation. The Ricci contribution in gradient form is:

$$\nabla_{\theta} J_{\text{Ricci}} = \sum_{(i,j) \in E} 2w_{ij}(S_i - S_j) \left(1 + \bar{\rho}_{ij} \right) \nabla_{\theta} \dots$$

with $\bar{\rho}_{ij} = \frac{|\rho_i| + |\rho_j|}{2}$.

Data and code availability

Simulation pipelines and figure-generation scripts are available within this project workspace and can be adapted to patient-specific manifold inputs.

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